

Thermal-Based Human Recognition via Spiking Neural Networks for Low-Power Embedded Systems

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Abstract

This research investigates privacy-preserving human-presence recognition from 32×24 thermal frames on a Raspberry Pi 5. A conventional convolutional neural network (CNN) and a spiking neural network (SNN) are trained on the same 1,800-frame dataset collected with an MLX90640 sensor. The CNN uses three convolutional blocks and two fully connected layers. The SNN retains the same trainable layer dimensions, adds Leaky Integrate-and-Fire neurons, repeatedly presents one static frame for 25 simulation steps, and classifies the frame from the output-layer membrane potential at the 25th step. On the random held-out test split of 270 frames, the CNN achieved an F1 score of 0.9912 and the SNN achieved 1.0000. A separate 600-frame evaluation, containing moving human samples and varied non-human thermal scenes, was completed for the CNN on both an RTX 4060 Ti and a Raspberry Pi 5. At the default threshold of 0.50, the CNN achieved an F1 score of 0.9117; the two platforms produced identical labels for all 600 samples. A threshold of 0.12 increased the F1 score to 0.9619 by trading additional false positives for fewer false negatives. These results demonstrate successful embedded deployment but also reveal a distribution gap between static training data and dynamic evaluation data. SNN evaluation on the separate 600 frames and controlled power measurement remain ongoing work.

1 Introduction

Human-presence sensing is required in smart environments, monitoring systems, assistive devices, and human–robot interaction. Visible-light cameras provide rich information, but they may reveal identity and appearance and may be affected by illumination. Low-resolution thermal array sensors instead record temperature distributions. At 32×24 pixels, the MLX90640 does not preserve the visual detail of an RGB camera, which makes it suitable for applications where privacy is important.

Embedded deployment introduces a second constraint: recognition must operate within limited computation and energy budgets. CNNs

are effective spatial feature extractors and provide a practical baseline for low-resolution thermal images. SNNs introduce membrane dynamics and spike-based activations and are often studied as a possible route toward energy-efficient inference. However, an SNN executed in a conventional PyTorch software loop is not automatically equivalent to sparse, asynchronous neuromorphic hardware. Therefore, energy efficiency must be established through measurement rather than assumed from the model type.

This study develops CNN and SNN classifiers using the same thermal dataset, deploys the trained models to a Raspberry Pi 5, and evaluates recognition accuracy, inference behavior, threshold selection, and platform consistency. The current research questions are: (1) whether low-resolution thermal frames can distinguish Human and No Human conditions across distance and posture; (2) whether GPU-trained models preserve their predictions after Raspberry Pi deployment; and (3) how CNN and SNN accuracy, latency, and measured energy compare under the same evaluation protocol.

2 Related Work

Thermal imaging has been used for human detection because it remains useful under poor illumination and can reduce the privacy exposure associated with visible imagery [1]. Earlier work demonstrated CNN-based people-presence detection on embedded platforms using low-resolution thermal images [2], while outdoor detection studies showed that very low spatial resolution creates challenges related to target scale and environmental variation [3]. Privacy-aware crowd counting and public low-resolution infrared datasets further established thermal arrays as a useful sensing modality for occupancy and activity analysis [4][5].

Recent work has also examined position estimation from low-resolution infrared arrays [6]. These studies primarily use conventional neural networks or task-specific pipelines. In parallel, surrogate-gradient methods have made multi-layer SNNs trainable with gradient-based optimization [7]. The present work connects these areas by comparing a CNN and an LIF-based SNN with matched trainable dimensions, identical thermal-frame preprocessing, and deployment on the same embedded platform. Its emphasis is not only the random test split but also the distribution shift observed in a separately collected evaluation set.

3 Methodology

3.1 System and Data Acquisition

The sensing platform consists of an MLX90640 thermal sensor connected to a Raspberry Pi 5 through I²C, as shown in Fig. 1. The sensor produces 768 temperature values arranged as a 24×32 frame. It was operated at a 4 Hz refresh rate with an 800 kHz I²C setting and positioned approximately 60 cm above the floor in an indoor laboratory. GPU training and evaluation used an NVIDIA GeForce RTX 4060 Ti with 16 GB memory. Figure 2 shows the overall research pipeline, consisting of thermal data collection, CNN and SNN training, deployment on the Raspberry Pi 5, and performance and power evaluation.

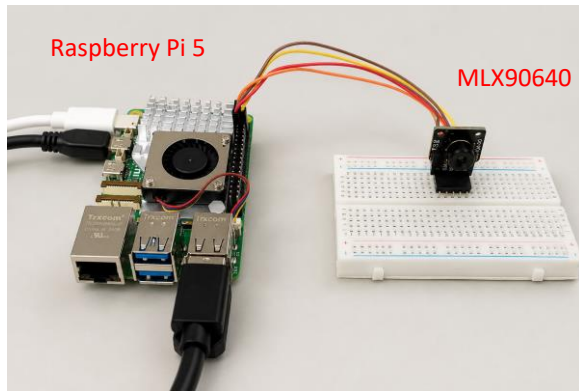


Figure 1. Experimental setup consisting of the MLX90640 thermal sensor and Raspberry Pi 5 connected via I²C.

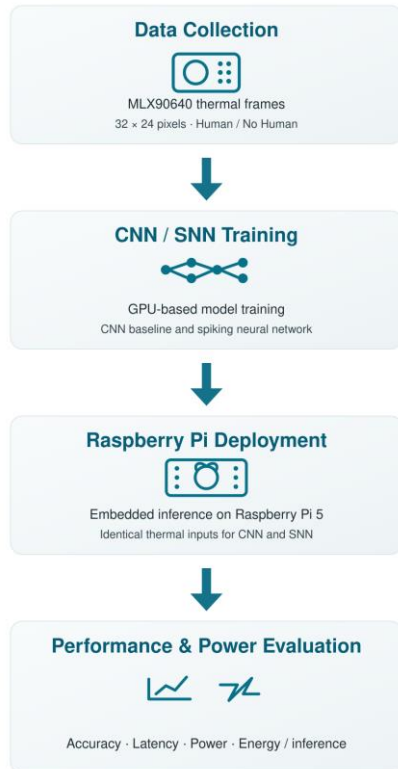


Figure 2. Overall pipeline of the proposed thermal-based human recognition system.

Each frame is independently normalized by $x' = (x - \min(x)) / (\max(x) - \min(x) + 10^{-6})$. This emphasizes the relative spatial temperature pattern rather than absolute temperature. The normalized frame is converted to a float32 tensor of shape $1 \times 1 \times 24 \times 32$. A probability at or above the selected threshold is classified as Human.

3.2 Dataset and Class Definitions

The training dataset contains 1,800 saved frames: 800 Human and 1,000 No Human. Human means that the target subject is present in the sensor field of view; No Human means that no person is present, although furniture, computers, equipment, hot objects, or cold objects may remain in the scene. Representative thermal frames from the Human and No Human classes are shown in Fig. 3. Human samples were collected from one subject, approximately 155 cm tall, at seven distances from 0.5 to 5.0 m and standing and sitting configurations. The dataset was shuffled with seed 42 and split without stratification into 70% training, 15% validation, and 15% test subsets.

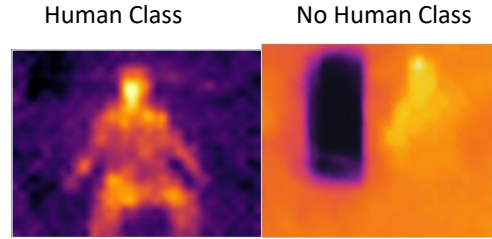


Figure 3. Representative thermal frames from the Human and No Human classes.

Split	Total	Human	No Human
Training	1,260	575	685
Validation	270	111	159
Test	270	114	156

Table 1. Random dataset split used for model development.

A separate set of 600 frames was collected after model development and was not part of the 1,800-frame random split. Its 300 Human frames were captured at 500 ms intervals while the subject moved at approximately 0.5–3.0 m and changed posture and orientation. Its 300 No Human frames contain varied laboratory backgrounds and thermal objects. The saved .npy files were presented in the same order to the GPU and Raspberry Pi. This evaluation is referred to as sequential saved-data inference; it does not include live sensor acquisition time.

3.3 CNN

The CNN contains three Conv–BatchNorm–ReLU–MaxPool blocks with 16, 32, and 64 output channels. The resulting $64 \times 3 \times 4$ feature map is flattened to 768 values, followed by a 128-unit fully connected layer with ReLU and dropout (0.5), and a one-unit output with sigmoid. The model has 122,081 trainable parameters. It was trained for up to 50 epochs with Adam (learning rate 0.001, weight decay

10^{-5}), binary cross entropy, batch size 32, and checkpoint selection by validation F1.

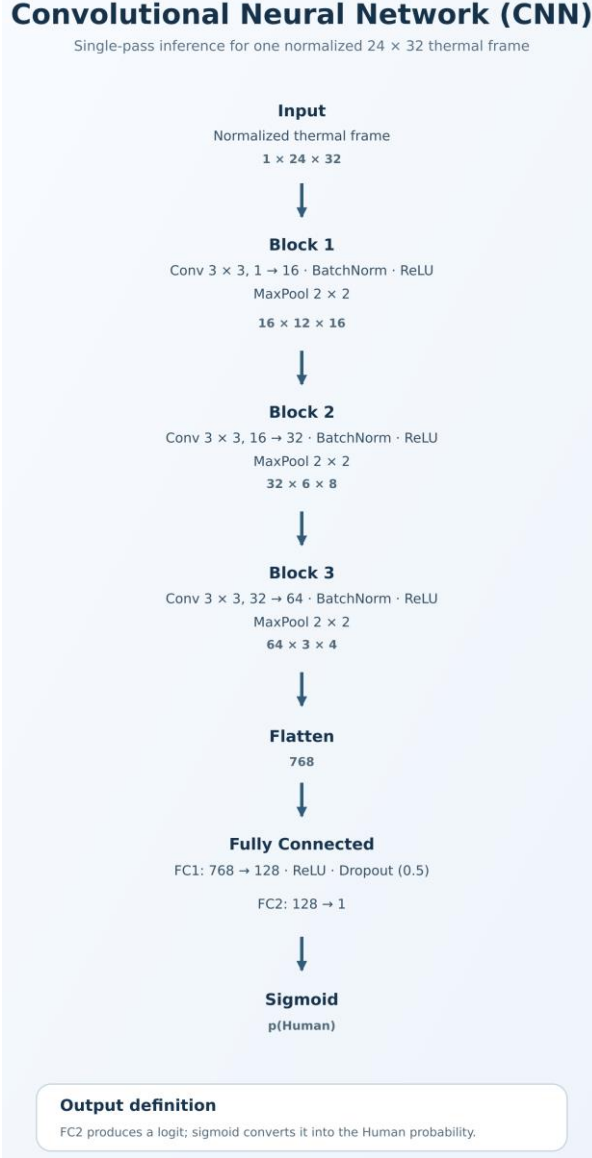


Figure 4. CNN architecture used in this study.

3.4 SNN

The SNN uses the same convolutional channel counts and fully connected dimensions as the CNN. Each convolutional stage follows Conv–BatchNorm–MaxPool–LIF, and LIF neurons also follow the first fully connected layer and the one-unit output layer. The same static thermal frame is repeated for $T = 25$ simulation steps; it is not a sequence of 25 different camera frames. The LIF decay factor is $\beta = 0.9$, and Fast Sigmoid surrogate gradients with slope 25 are used.

The classification logit is the output-layer membrane potential at the 25th step. Sigmoid is applied to this membrane potential to obtain the Human probability. Output spikes and firing rate are auxiliary statistics only and do not determine the class. Training uses BCEWithLogitsLoss, Adam, and the same learning rate, weight decay, batch size, and maximum epoch count as the CNN.

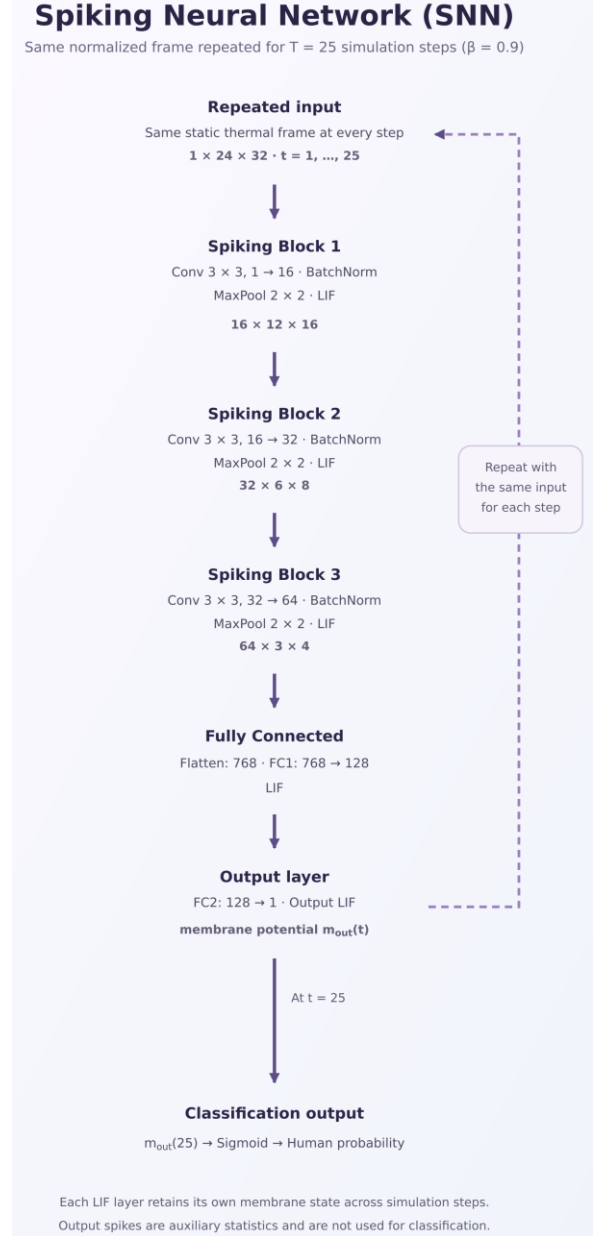


Figure 5. SNN architecture and membrane-potential output.

4 Experimental Evaluation

4.1 Random-Split Static Test

The held-out random test subset contains 270 frames. The CNN misclassified two Human frames and no No Human frames, producing an accuracy of 0.9926 and an F1 score of 0.9912. The SNN classified all 270 frames correctly. Because these frames originate from the same collection procedure as the training data, this result measures performance within a closely matched distribution and does not by itself establish robustness to moving subjects or new environments.

Model	Accuracy	Precision	Recall	F1
CNN	0.9926	1.0000	0.9825	0.9912
SNN	1.0000	1.0000	1.0000	1.0000

Table 2. Static test results on 270 held-out frames.

4.2 CNN Evaluation on Separate Data

At threshold 0.50, the CNN produced 253 true positives, 47 false negatives, 298 true negatives, and two false positives over the separate 600 frames. The combined accuracy was 0.9183, precision 0.9922, recall 0.8433, and F1 score 0.9117. The lower recall relative to the random test split is associated with moving subjects and continuous changes in distance, posture, and orientation. The two false positives contained hot objects with maximum temperatures of approximately 68.48 °C and 73.11 °C, suggesting that localized high-temperature regions can resemble human thermal features.

The RTX 4060 Ti and Raspberry Pi 5 generated the same binary prediction for all 600 saved frames using the same weights and threshold. The average absolute probability difference for the 300 Human frames was approximately 4.41×10^{-5} , with a maximum of approximately 1.96×10^{-4} . Thus, numerical differences between platforms did not alter any label.

Platform	Class	Outcome	Median time
RTX 4060 Ti	Human	TP 253 / FN 47	1.773 ms
RTX 4060 Ti	No Human	TN 298 / FP 2	1.762 ms
Raspberry Pi 5	Human	TP 253 / FN 47	1.189 ms
Raspberry Pi 5	No Human	TN 298 / FP 2	1.209 ms

Table 3. CNN sequential saved data evaluation. Timing includes model forward execution only.

The timing values should not be interpreted as a strict GPU-versus-Raspberry-Pi speed comparison. File loading, normalization, tensor construction, device transfer, resource monitoring, visualization, and sensor acquisition were excluded. In addition, the GPU script used wall-clock timing without explicit CUDA synchronization, so a controlled repeated benchmark is required for rigorous latency comparison.

4.3 Threshold Analysis

The output threshold represents the minimum predicted Human probability required for a positive decision. The default value of 0.50 is a conventional operating point, not a value learned from application costs. A sweep on the separate 600 frames found the maximum F1 score at 0.12. This reduced false negatives from 47 to 10 but increased false positives from two to 13. A threshold of 0.11 produced recall 0.9700 while keeping the false-positive rate at 0.0500. Therefore, the final threshold should be selected according to whether missed detections or false alarms are more costly in the target application.

Threshold	TP	FN	FP	Precision	Recall	F1
0.50	253	47	2	0.9922	0.8433	0.9117
0.12	290	10	13	0.9571	0.9667	0.9619
0.11	291	9	15	0.9510	0.9700	0.9604

Table 4. CNN threshold trade-off on the separate 600-frame set.

5 Discussion

The results support three current conclusions. First, the CNN and SNN can separate Human and No Human frames in the random test split. Second, the CNN model can be transferred from the GPU environment to the Raspberry Pi without changing its decisions on the 600 saved frames. Third, evaluation on independently collected, dynamic Human data is substantially more informative than the near-perfect random-split result.

Several limitations prevent broader claims. Data were collected indoors in one laboratory and Human data came from one subject. Random frame-level splitting may place highly similar samples in training and test subsets. Per-frame min-max normalization discards absolute temperature information and may amplify scene-relative patterns. The CNN threshold sweep was performed on the same 600 frames used to report performance and therefore needs validation on another held-out set before being treated as a final operating point. Finally, no controlled power measurement has yet demonstrated that the SNN is more energy efficient than the CNN on the Raspberry Pi.

6 Progress and Future Work

Data collection, preprocessing, CNN and SNN training, random-split testing, CNN deployment, the separate CNN evaluation, and GPU-Raspberry-Pi consistency analysis are complete. The remaining experimental work consists of evaluating the SNN on the separate 600-frame dataset and measuring the power consumption of the CNN and SNN on the Raspberry Pi 5. Both models will be evaluated under the same input and measurement conditions to enable a fair comparison of recognition performance, inference time, power consumption, and energy per inference. The final results will then be summarized in the master's thesis.

Period	Planned work	Completion goal
Aug – Sep 2026	SNN inference evaluation	Accuracy and latency comparison
Oct – Nov 2026	Power measurement	Power and energy comparison
Dec 2026 – Feb 2027	Thesis preparation	Thesis completion

Table 5. Dated research plan toward thesis completion.

7 Conclusion

This work establishes an embedded pipeline for Human/No Human recognition from 32×24 thermal frames. Both models achieved high accuracy on the random held-out split, and the CNN produced identical decisions on an RTX 4060 Ti and Raspberry Pi 5 for all 600 separately collected frames. However, CNN recall decreased under dynamic Human conditions, demonstrating that a random frame split can overestimate real-world robustness. Threshold adjustment improved the balance between missed detections and false alarms, but the operating point remains application dependent. The next decisive steps are to complete the same separate-data evaluation for the SNN and to measure power and energy under a controlled protocol. Until those measurements are complete, the SNN should be described as a spike-based multi-step model under investigation, not as a demonstrated low-power solution.

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